

Reproduction of hiyouga/LlamaFactory: Qwen3-4B LoRA supervised fine-tuning example

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Independent reproduction · Artifact evaluation · Natural language processing

Abstract

Ran the shipped example `examples/train_lora/qwen3_lora_sft.yaml` at the pinned commit unchanged except for the local model path and output directory: LoRA (rank 8, all linear layers) supervised fine-tuning of Qwen3-4B-Instruct-2507 on the bundled identity and alpaca_en_demo datasets (up to 1,000 samples each), 3 epochs, batch 1 with 8 accumulation steps, learning rate 1e-4 cosine, bf16. torchaudio was replaced with the CUDA 12.8 build matching PyTorch. One NVIDIA RTX 5060 Ti.

About this record

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Qwen3-4B LoRA SFT, 3 epochs	2026-10-08	1	1	Signed by the K-Veritas server

2 Results

r1: Qwen3-4B LoRA SFT, 3 epochs

Run	Metric	Final value	Step
1	train_loss	0.849146	410
1	train_runtime_s	1181	
1	train_samples_per_second	2.769	

Run	Inline claim	Value	Phase
1	train_loss	0.999387	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	1194.0 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	5141f45aee02eb7c77ff91f7ac9372011689fbe25d0605221ead4a17ced170d8	7.1 MB
r1.kvbundle.zip	0a680f8869eee4c0bcad32ae50d68487a9b387ee0a0bd47537b0f6c8a9da5586	2.8 MB

5 Links and licence

Paper: <https://arxiv.org/abs/2403.13372>

Reproduction of: <https://github.com/hiyouga/LlamaFactory/tree/ce9dc9e072f80fa3abe0989d4ab90da25f083438>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: lora, sft, qwen3, fine-tuning.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Qwen3-4B LoRA SFT, 3 epochs), kv:2610.00026v1/r1. Data hash f8c48c4d67d199557674230f0069a6339f880f9716e54532e583e14e3bbfd2a5.

<https://kveritas.org/records/2610.00026v1/r1.pdf>